# Ionic Compounds

Ionic compounds are formed from a positive ion and a negative ion combining together. The ions combine together so that the positive charges and the negative charges balance each other out to form a neutral compound.

Table salt is a common ionic compound. It is formed when one sodium ion, **$Na^+$**, combines with one chloride ion, **$Cl^-$**. The positive charge on the sodium ion balances out the negative charge on the chloride ion to form **sodium chloride, NaCl**.

**Your Task:** You will be given a set of wooden blocks representing different ions. The length of each block corresponds to the size of the charge on the ion. One colour will represent the positive ions and a second will represent the negative ions. Use the wooden blocks to construct models of the ionic compounds that can be formed. Once you have constructed a compound, fill in the corresponding row of the table.

| Compound | Diagram of Positive Ion | Diagram of Negative Ion | Diagram of Compound | Compound Formula | Total Positive Charge | Total Negative Charge | Overall Charge of Compound |
|---|---|---|---|---|---|---|---|
| Sodium chloride | | | | | | | |
| Sodium oxide | | | | | | | |
| Sodium nitride | | | | | | | |

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|---|---|---|---|---|---|---|---|
| Magnesium chloride | | | | | | | |
| Magnesium oxide | | | | | | | |
| Magnesium nitride | | | | | | | |
| Aluminium chloride | | | | | | | |
| Aluminium oxide | | | | | | | |
| Aluminium nitride | | | | | | | |
| Lead (IV) chloride | | | | | | | |
| Lead (IV) oxide | | | | | | | |

Use the blocks to write the formula of the following compounds:

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|---|---|---|---|
| Lithium chloride fluoride | Calcium bromide | Barium sulphide | Aluminium |
| Calcium chloride sulphide | Lead (IV) sulphide | Potassium oxide | Aluminium |